

# Lattice Green's function of the hyperkagome lattice: modular uniformization at level 30 from an orthogonal differential Galois group

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The lattice Green's functions of the cubic lattices are classical: each is a symmetric square of a second-order operator, with a closed form in complete elliptic integrals. The hyperkagome lattice, realized by the iridium sublattice of the spin-liquid candidate  $\text{Na}_4\text{Ir}_3\text{O}_8$ , has resisted such a treatment: Varma and Monien reduced its density of states to a threefold integral they found no way to solve exactly. From exact lattice moments we obtain an irreducible third-order linear differential (Picard–Fuchs) operator annihilating the Green's function. It is not a literal symmetric square, yet its symmetric square carries a rational solution: the differential Galois group is orthogonal, and an order-two differential intertwiner makes the operator projectively a symmetric square. Our main result is that the underlying second-order operator uniformizes the genus-zero modular curve of the level-thirty Fricke group, an explicit eta quotient parametrizing the natural variable; the Schwarzian identity is proved by a pole-degree bound, not checked numerically. The hyperkagome Green's function is therefore modular at level thirty, supplying the closed form Varma and Monien sought. Its weight-two period is obtained explicitly: a depth-one quasimodular form provably not a modular form times an algebraic function at any weight.

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## 1. Introduction

The lattice Green’s function (LGF)

$$(1) \quad G(z) = \int_{\text{BZ}} \frac{d^3k}{(2\pi)^3} \frac{1}{z - \varepsilon(\mathbf{k})} = \int \frac{\rho(E) dE}{z - E},$$

the resolvent of a tight-binding band  $\varepsilon(\mathbf{k})$  averaged over the Brillouin zone [3], is a central object of lattice statistics: it controls random-walk return probabilities, defect and impurity states, and the local density of states  $\rho(E) = -\pi^{-1} \text{Im } G(E+i0^+)$  [13, 16]. For the three-dimensional cubic lattices the LGF is a celebrated closed form. Watson’s triple integral for the simple-cubic (SC) lattice [34], and the body- and face-centered analogues, are all expressible through products of complete elliptic integrals  $K$  [19, 15, 20]; structurally, each of these order-three linear differential operators is the *symmetric square* of a second-order (elliptic) one [9], which is precisely why an elliptic closed form exists.

Not every lattice is so fortunate. As the local geometry becomes more intricate—decorated, frustrated, or multi-band lattices—the LGF can fail to reduce to elliptic integrals and instead becomes a period of a higher-order or irreducible linear differential operator (the diamond lattice already signals this trend [16, 20]). The natural three-dimensional frustrated network is the *hyperkagome* lattice: a chiral arrangement of corner-sharing triangles, obtained by removing one quarter of the sites of the pyrochlore lattice in an ordered (space group  $P4_132$ ) fashion. It is realized by the  $\text{Ir}^{4+}$  moments of  $\text{Na}_4\text{Ir}_3\text{O}_8$ , a leading three-dimensional quantum-spin-liquid candidate [30, 18, 24], which has made its single-particle spectral functions of independent interest.

Varma and Monien [33] computed the hyperkagome density of states and reduced it to a single threefold integral of a rational function of  $\cos k_x, \cos k_y, \cos k_z$ , remarking that “it seems highly suggestive that a closed form expression can be obtained if a particular complicated yet highly symmetrical integral can be solved,” but that “we have currently found no way to exactly solve” it. That integral has remained open.

We settle its analytic structure and prove its modularity. The hyperkagome LGF is governed by an *irreducible third-order* linear differential (Picard–Fuchs) operator  $M$  that is not a *literal* symmetric square, yet whose symmetric square carries a rational solution: the differential Galois group is *orthogonal*,  $G_M = \text{O}(3, \mathbb{C})$  with  $G_M^\circ = \text{SO}(3, \mathbb{C}) \cong \text{PSL}(2, \mathbb{C})$ , so  $M$  is projectively equivalent to the symmetric square of a second-order operator. The obstruction Varma and Monien met is therefore not a genuine third-order

(“ $\text{SL}_3$ -generic”) object with no elliptic reduction; it is a *hidden orthogonal / modular structure*, realized through a nontrivial operator equivalence rather than the function-multiplier and pullback equivalences one usually tests. We obtain the operator explicitly, certify this structure by exact computation, and thereby explain *why* a naive symmetric-square reduction was never found. We then prove the closed-form structure outright: the second-order operator underlying  $M$  is the uniformizing operator of the genus-zero modular curve  $X(\Gamma_0(30)^+)$ , the natural variable  $t$  generates its function field and is given by an explicit level-30 eta-quotient parametrization, and the hyperkagome LGF is modular at level  $30 = 2 \cdot 3 \cdot 5$ —to our knowledge the first three-dimensional LGF known to require such an equivalence, and the first lattice Green’s function realized at a modular level with three distinct prime factors.

## 2. Hyperkagome lattice and its Green’s function

The hyperkagome lattice has twelve sites per cubic cell, each with coordination number four; its nearest-neighbor tight-binding Bloch Hamiltonian  $H(\mathbf{k})$  is a  $12 \times 12$  Hermitian matrix. We fix the hopping amplitude to unity (reserving  $t$  for the spectral variable introduced below) and the energy origin at the band center. Diagonalizing  $H(\mathbf{k})$  over the Brillouin zone yields the spectrum shown in Figure 1. Three exact features organize everything that follows.

(i) *A flat band, and the line-graph structure.* The hyperkagome lattice is the line graph of a three-regular “premedial” net  $\Gamma$  (eight vertices per cell, twelve edges): its sites are the edges of  $\Gamma$  and its triangles are the vertices of  $\Gamma$ . For the line graph  $L(\Gamma)$  of a  $d$ -regular graph one has  $A(L(\Gamma)) = B^\top B - 2I$  with  $B$  the incidence matrix, so the spectrum is  $\{\mu_i + d - 2\}$  together with the eigenvalue  $-2$  of multiplicity (edges)  $-(\text{vertices})$ . Here  $d = 3$ , giving four flat bands at  $E = -2$  (spectral weight exactly  $1/3$ , i.e.  $\frac{12-8}{12}$ ); this contributes a pole  $\frac{1}{3}(z+2)^{-1}$  to  $G(z)$  and is the flat-band pole written  $1/(t_{\text{VM}}+1)$  in [33], whose energy variable is  $t_{\text{VM}} = E/2$  (the correspondence, including this pole and its weight, is reproduced in their variable by `numerics/vm_crosscheck.py` in the repository). The eight dispersive bands are  $E = \mu_i + 1$  and fill  $[-2, 4]$ , since the adjacency eigenvalues  $\mu_i$  of a three-regular graph lie in  $[-3, 3]$ .

(ii) *An exact reflection symmetry, from bipartiteness.* A graph has an adjacency spectrum symmetric about 0 if and only if it is bipartite; we verify that the premedial net  $\Gamma$  is bipartite (it admits a translation-consistent two-coloring, checked on finite blocks up to  $5^3$  cells). Through the line-graph shift  $E = \mu + 1$  this makes the dispersive density of states *exactly* symmetric about  $E = 1$ ; independently, all odd central moments  $\int \rho_{\text{disp}}(E)(E - 1)^{2m+1} dE$

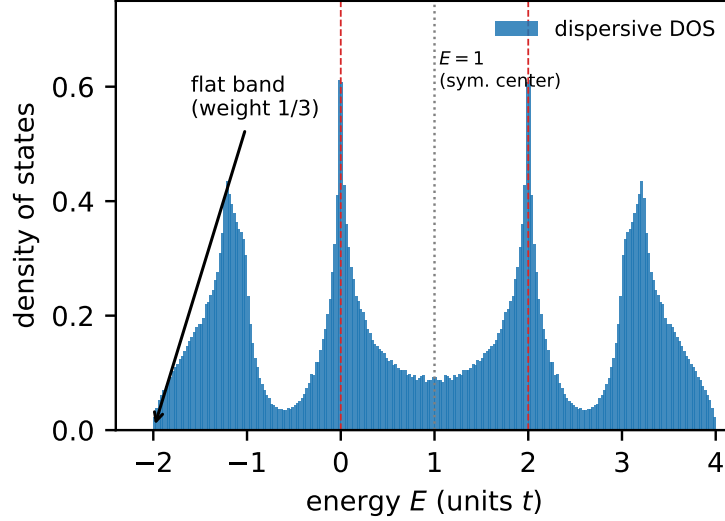


Figure 1: Density of states of the hyperkagome tight-binding model at unit hopping. Four flat bands at  $E = -2$  carry weight  $1/3$ ; the eight dispersive bands fill  $[-2, 4]$  and are *exactly* symmetric about  $E = 1$ , with principal van Hove singularities at  $E = 0, 2$  (dashed) and weaker interior features at  $E = -1, 3$  and  $E = 1 \pm \sqrt{5}$  (see text).

vanish in exact arithmetic for every odd order up to 229, the highest the tabulated moments support. Hence the dispersive part of  $G$  is odd in  $\zeta \equiv z - 1$ , and is a function of  $t \equiv \zeta^{-2}$ .

(iii) *Integer moments.* The per-site spectral moments  $m_n = \frac{1}{12} \text{Tr} H^n$  are exact integers,  $m_0, \dots, m_{10} = 1, 0, 4, 4, 28, 60, 260, 756, 2828, 9292, 33384$ ; we computed them to  $m_{230}$  by closed-walk enumeration. Among the ordered quarter-depletions of the pyrochlore lattice, the chiral hyperkagome enantiomorphs are the ones reproducing this sequence (hexagon-containing depletions already differ at  $m_6 = 264 \neq 260$ ); this selection is checked separately and is not needed for what follows.

### 3. The Picard–Fuchs operator

Because  $G(z)$  is the diagonal of a rational function over the torus, it is holonomic (D-finite): it satisfies a linear differential equation with polynomial coefficients [27]. We determine that equation from the moments. Throughout, *operator* abbreviates *linear differential operator* in  $d/dt$  with coefficients in  $\mathbb{Q}(t)$ ,

and all products, factorizations, adjoints and symmetric squares are taken in the ring  $\mathbb{Q}(t)\langle d/dt \rangle$  of such operators. Removing the flat band,  $d_n = m_n - \frac{1}{3}(-2)^n$ , and forming the even central moments  $\nu_m = \int \rho_{\text{disp}}(E) (E-1)^{2m} dE$ , the generating function  $\Phi(t) = \sum_m \nu_m t^m$  (with  $t = \zeta^{-2}$ ) is the essential object. Guessing a minimal annihilator from the exact  $\nu$  sequence by the method of differential approximants [16, 21] returns a linear differential operator  $L_4$  of order four and degree fifteen.

This  $L_4$  is reducible in a trivial way: it admits the constant solution, so  $L_4 = M d/dt$  exactly, with  $M$  of order *three* and degree fifteen,

$$(2) \quad M = \sum_{i=0}^3 c_i(t) \left( \frac{d}{dt} \right)^i, \quad \deg c_i \leq 15,$$

the minimal annihilator of  $\Phi'$ —minimal in the certified, guess-and-verify sense made precise under *Certification* below. That factorization is moreover the *only* one:  $L_4$  is not a least common left multiple  $\text{LCLM}(N, d/dt)$  of  $d/dt$  with a second operator of order three, so the constant solution is not a direct summand and  $M$ —rather than some other order-three operator—is the intrinsic object here (Proposition 4.1 below). Its leading coefficient factorizes as

$$(3) \quad c_3(t) \propto t^2 (t - \tfrac{1}{9})(t - \tfrac{1}{5})(t - \tfrac{1}{4})(t - 1)^3 p_7(t),$$

with  $p_7$  an irreducible degree-seven polynomial. The finite singular points are images of the band structure under  $t = (E - 1)^{-2}$ : the band edge  $E = 4$  (equivalently  $E = -2$ ) maps to  $t = \frac{1}{9}$ , the principal van Hove energies  $E = 0, 2$  to  $t = 1$ , and the interior van Hove features  $E = -1, 3$  and  $E = 1 \pm \sqrt{5}$  to  $t = \frac{1}{4}$  and  $t = \frac{1}{5}$ ; the half-integer exponent in (4) confirms a square-root singularity at each. The remaining singular points ( $t = 0$ ,  $t = \infty$ , and the degree-seven locus  $p_7$ ) all appear in the complete Riemann scheme (4) below.

*Certification.* The operator  $M$  was reconstructed exactly over  $\mathbb{Q}$  and shown to annihilate the series  $\Phi'(t)$  *exactly over*  $\mathbb{Q}$  through order  $t^{111}$ —112 independent relations against the 57 free coefficients of  $M$  (one of the 58 stored coefficients fixes the overall scale), an overdetermination margin of 55 with zero residuals. It was furthermore re-derived from a second, independent implementation by the authors—an independent quarter-depletion construction, an independent closed-walk moment enumeration, and an independent guess—and the moment sequence and all four coefficient polynomials of  $M$  came out bit-identical. As a further independent check, the integer moments were reproduced by direct diagonalization of the  $12 \times 12$  Bloch Hamiltonian

on a Brillouin-zone grid—a method disjoint from the closed-walk enumeration. This is the standard “conjectured-then-verified” certification of the lattice-statistics literature [16]. An unconditional derivation by creative telescoping [22, 8] from the rational integrand of the resolvent—which would upgrade this verification to a formal proof—is a natural next step, well suited to a dedicated holonomic-functions implementation.

#### 4. Riemann scheme and differential-Galois structure

The analytic nature of the LGF is decided by the local structure of  $M$ . Computing the Frobenius indicial exponents at each regular singular point gives the Riemann scheme

$$(4) \quad \begin{array}{c|ccccccc} & t=0 & t=\frac{1}{9} & t=\frac{1}{5} & t=\frac{1}{4} & t=1 & p_7 & t=\infty \\ \hline \text{exp.} & \{-1, -1, 0\} & \{-\frac{1}{2}, 0, 1\} & \{-\frac{1}{2}, 0, 1\} & \{-\frac{1}{2}, 0, 1\} & \{-1, -\frac{1}{2}, \frac{1}{2}\} & \{0, 1, 3\} & \{\frac{3}{2}, 2, 3\} \end{array}$$

Here  $p_7$  denotes the irreducible degree-seven locus, whose seven Galois-conjugate roots all carry the same integer exponents  $\{0, 1, 3\}$ , computed exactly by reduction modulo  $p_7$ . These are *apparent* singularities: integer exponents alone do not establish this, but all three local solutions at each root are log-free—indeed analytic. Log-freeness is verified directly through the Frobenius obstruction at each root of  $p_7$ , computed modulo several independent primes; independently of that check, analyticity follows unconditionally from the intertwiner of Section 7: the exact operator identity of step [IV] of Theorem 7.1, together with the injectivity computation (18), carries a full solution basis of the operator  $\tilde{N}$ —which is regular on the  $p_7$  locus, as are the twist unit  $v$  and the first-order operator  $T$  of (15)—onto a basis of solutions of  $M$ , so every local solution of  $M$  at each root of  $p_7$  is analytic. We use the standard Riemann-scheme convention at infinity, in which an exponent  $\rho$  at  $t = \infty$  corresponds to a local solution  $y \sim t^{-\rho}$ ; the three solutions there thus behave as  $t^{-3/2}$ ,  $t^{-2}$  and  $t^{-3}$ . As a check on (4), the operator has  $s = 13$  singular points—the five rational points above, the seven roots of  $p_7$ , and  $t = \infty$ ; the relation counts the apparent singularities on  $p_7$  along with the rest—and the exponents sum to

$$(5) \quad -2 + 3 \cdot \frac{1}{2} - 1 + 7 \cdot 4 + \frac{13}{2} = 33 = \frac{n(n-1)}{2} (s - 2), \quad n = 3,$$

so the Fuchs relation is satisfied exactly.

The repeated exponent  $-1$  at  $t = 0$  is the source of the logarithmic solution used below. The half-integer exponent  $-\frac{1}{2}$  at the band edge  $t = \frac{1}{9}$  is the square-root van Hove singularity of a three-dimensional band edge,  $\rho(E) \sim \sqrt{E_{\text{edge}} - E}$ .

We first record two local-to-global facts about  $M$ —it is irreducible, and it is not a *literal* symmetric square—and then determine its differential Galois group, which is the object that actually decides the closed-form question.

$M$  is irreducible over  $\overline{\mathbb{Q}}(t)$ .  $M$  is Fuchsian—regular singular at every point, including  $t = \infty$  (exponents  $\{\frac{3}{2}, 2, 3\}$ )—so any first-order right factor  $d/dt - h$  has  $h = \sum_s e_s/(t - s)$  with  $e_s$  a local exponent at each singular point  $s$  and *no* polynomial part. At the rational singular points the  $e_s$  are the exponents of (4); at the irreducible degree-seven locus  $p_7$  the common (Galois-invariant) residue ranges over its exponents  $\{0, 1, 3\}$ . This candidate set is finite, and each candidate is tested exactly over  $\mathbb{Q}$  by the divisibility criterion  $c_3(h'' + 3hh' + h^3) + c_2(h' + h^2) + c_1h + c_0 = 0$ . No first-order right factor of  $M$  exists over  $\mathbb{Q}(t)$ , and none of  $\text{adjoint}(M)$ —which for a third-order operator excludes an order-two right factor of  $M$  as well—so  $M$  has no right factor of order one or two over  $\mathbb{Q}(t)$ . We now descend to  $\overline{\mathbb{Q}}(t)$  in two steps. *First*,  $M$  has no order-one right factor there: a first-order right factor  $d/dt - h$  with  $h \in \overline{\mathbb{Q}}(t)$  has a finite Galois orbit  $\{d/dt - h^\sigma\}$  of right factors of  $M$ , whose least common left multiple is a right factor *defined over*  $\mathbb{Q}(t)$  of order  $k \in \{1, 2, 3\}$ ;  $k = 1$  gives  $h \in \mathbb{Q}(t)$  and  $k = 2$  an order-two right factor over  $\mathbb{Q}$ , both excluded above, while  $k = 3$  would make  $M$  the LCLM of order-one operators, hence completely reducible, forcing *semisimple* local monodromy and no logarithm—contradicting the genuine logarithm at  $t = 0$  (repeated indicial exponent  $-1$ , exactly one log-free local solution of three, verified directly). *Second*,  $M$  has no order-two right factor over  $\overline{\mathbb{Q}}(t)$ : such a factor is equivalent to an order-one right factor of  $\text{adjoint}(M)$  over  $\overline{\mathbb{Q}}(t)$ , excluded by the identical descent applied to  $\text{adjoint}(M)$  (no order-one factor over  $\mathbb{Q}$  by the adjoint search, no order-two by the  $M$  search, and  $\text{adjoint}(M)$  is not completely reducible, since complete reducibility is self-dual and  $M$  is not). Hence  $M$  is irreducible over  $\overline{\mathbb{Q}}(t)$ . (This exact enumeration replaces the floating-point candidate search of earlier drafts; the computation is carried out by `numerics/certify_factor.py` in the repository and is independently reproducible with a certified factorizer such as Maple's `DFactor` or Magma.)

Irreducibility settles a second question, raised already in Section 3: an order-four operator with a constant solution admits the factorization  $L_4 = M d/dt$ , but it may instead be the *direct sum*  $\text{LCLM}(N, d/dt) = N \oplus d/dt$ , in which case  $N$ —not  $M$ —is the operator to study, and every computation below would have to be redone for it. That does not happen here.

**Proposition 4.1.**  *$L_4 = M d/dt$  is the unique factorization of  $L_4$  into linear differential operators of positive order. In particular  $L_4 \neq \text{LCLM}(N, d/dt)$  for every order-three operator  $N$ : the constant solution is not a direct summand of  $\text{Sol}(L_4)$ .*

*Proof.* Factorizations of  $L_4$  correspond to submodules of  $\text{Sol}(L_4)$ , and  $d/dt$  maps  $\text{Sol}(L_4)$  onto  $\text{Sol}(M)$  with kernel the constants  $\mathbb{C}$ . Since  $M$  is irreducible, a one-dimensional submodule has image 0 and therefore lies in  $\mathbb{C}$ , while a two-dimensional one would have an image of dimension one or two—in either case a proper nonzero submodule of  $\text{Sol}(M)$ . So besides 0,  $\mathbb{C}$  and  $\text{Sol}(L_4)$  the only possible submodule is a three-dimensional  $S$ ; such an  $S$  meets  $\mathbb{C}$  trivially (otherwise its image would again be two-dimensional), giving  $\text{Sol}(L_4) = S \oplus \mathbb{C}$ , i.e. exactly the LCLM case with  $\text{Sol}(N) = S$ .

Suppose such an  $S$  existed. Then  $d/dt: S \rightarrow \text{Sol}(M)$  is an isomorphism of differential modules, and it restricts to one of the local solution spaces on a punctured disc around  $t = 1$ , where we now work. There the exponents of  $M$  are  $\{-1, -\frac{1}{2}, \frac{1}{2}\}$  by (4): the exponent  $-1$  is a simple root of the indicial equation and no exponent equals  $-1 + k$  for a positive integer  $k$ , so the Frobenius solution belonging to it carries no logarithm. Being a Laurent series in integer powers it is single-valued, and normalized it reads  $y = (t-1)^{-1}(1 + O(t-1))$ , of residue 1. Let  $Y \in S$  be its preimage and let  $\gamma$  be the local monodromy at  $t = 1$ , so that  $\gamma y = y$ . As  $\gamma$  lies in the monodromy group, hence in the differential Galois group, and  $S$  is a submodule,  $\gamma Y - Y \in S$ , while  $(\gamma Y - Y)' = \gamma y - y = 0$  puts it in  $\mathbb{C}$ ; as  $S \cap \mathbb{C} = 0$  we get  $\gamma Y = Y$ . But  $Y' = y$  forces  $Y = \log(t-1) + (\text{single-valued})$ , so  $\gamma Y = Y + 2\pi i$ —a contradiction.  $\square$

A second, global proof runs through the adjoint and is independent of the local one. Write  $L_4 = A N$  with  $N$  a right factor of order three and  $A$  of order one; since  $\text{adjoint}(A N) = \text{adjoint}(N) \text{adjoint}(A)$ , an order-three right factor of  $L_4$  is the same thing as an order-one right factor of  $\text{adjoint}(L_4)$ , i.e. a hyperexponential solution of it. Now  $\text{adjoint}(L_4) = -(d/dt) \text{adjoint}(M)$ , so such a solution obeys  $\text{adjoint}(M)(y) = c$  for a constant  $c$ . Applying an operator with rational coefficients to a hyperexponential  $y$  returns  $y$  times a rational function,  $\text{adjoint}(M)(y) = S y$  with  $S \in \overline{\mathbb{Q}}(t)$ : hence  $c = 0$  would make  $y$  a hyperexponential solution of the irreducible  $\text{adjoint}(M)$ , while  $c \neq 0$  gives  $y = c/S$ , a *rational* solution of  $\text{adjoint}(L_4)$ . There is none—and it is enough to search over  $\mathbb{Q}(t)$ , the space of rational solutions being Galois-stable, hence defined over  $\mathbb{Q}$ : the local exponents bound the denominator and the exponents at infinity bound the degree, and the resulting finite linear system over  $\mathbb{Q}$  has only the zero solution. Both proofs are certified in `numerics/certify_lclm.py` (`CERTIFICATE_lclm.txt`), which first validates its Frobenius and rational-solution primitives on operators of known structure—including a manufactured  $\text{LCLM}(N, d/dt)$ , on which the same search does find the rational solution, so the negative result is not vacuous.



*M is not a literal symmetric square.* If  $M$  were literally the symmetric square of a second-order operator with local exponents  $\{a, b\}$ , its three exponents would be  $\{2a, a + b, 2b\}$ , an arithmetic progression whose middle term is the mean of the outer two. No triple in (4) is of this form: at  $t = \frac{1}{9}$  the exponents  $\{-\frac{1}{2}, 0, 1\}$  have mean of outer terms  $\frac{1}{4} \neq 0$ , and at  $t = 1$  the exponents  $\{-1, -\frac{1}{2}, \frac{1}{2}\}$  have mean of outer terms  $-\frac{1}{4} \neq -\frac{1}{2}$ ; a single such point already suffices. As a positive control, the same analysis applied to the simple-cubic LGF returns the symmetric-square-compatible exponents  $\{0, \frac{1}{2}, 1\} = \{2a, a + b, 2b\}$  with  $\{a, b\} = \{0, \frac{1}{2}\}$ , correctly reflecting its closed form in  $K$ . *The scope of this exclusion is limited, and it is essential not to overstate it.* A gauge transformation shifts the three exponents at each point by a common constant, and an algebraic pullback multiplies them by the local ramification index; both preserve arithmetic progressions, so the exponent test also rules out every *function-multiplier or pullback* equivalent of a symmetric square. It does *not*, however, rule out an equivalence realized by a *differential intertwiner*—a homomorphism of differential modules of order two, which shifts the individual local exponents by *different* integers and so destroys the arithmetic progressions while preserving the Galois-theoretic content. This is exactly the loophole the hyperkagome operator exploits: as we show in Section 5,  $M$  is equivalent to a symmetric square through such an order-two intertwiner, even though it is not a literal symmetric square (nor a multiplier/pullback of one). Correcting an overstatement in an earlier version of this work, the not-a-literal-symmetric-square property therefore does *not* exclude an elliptic closed form; the closed-form question is settled instead by the Galois group, which we now compute.

*No algebraic or elementary closed form.* On a separate axis,  $M$  has no *Liouvillian* solutions [32], hence no closed form in algebraic or elementary functions. By the Singer–Ulmer classification [31], an irreducible third-order operator has Liouvillian solutions only if its differential Galois group is finite or imprimitive. A finite group makes every solution algebraic, hence log-free. An imprimitive group permutes three lines, so in the corresponding basis every element—in particular every local monodromy, since for a Fuchsian operator the Galois group is the Zariski closure of the monodromy group—is a monomial matrix. Such a matrix is necessarily semisimple: if its permutation part has order  $k$ , its  $k$ -th power is diagonal and invertible, and an invertible matrix with a diagonalizable power is itself diagonalizable in characteristic zero. In either case there can be *no logarithmic solution*. But  $M$  has a genuine logarithmic solution: at  $t = 0$  the indicial exponent  $-1$  is repeated and only one log-free (Laurent) local solution exists (of three), so the local monodromy is non-semisimple. This excludes both the finite and the imprimitive

cases, so the Galois group acts irreducibly and non-solvably and  $M$  is non-Liouvillian. The differential-Galois computation of the next section gives the same conclusion more sharply:  $G_M^\circ = \mathrm{SO}(3, \mathbb{C})$  is simple, hence non-solvable, and an irreducible operator has Liouvillian solutions if and only if  $G_M^\circ$  is solvable. Being non-Liouvillian does *not* preclude an elliptic form—the complete elliptic integral  $K$  is itself non-Liouvillian, and the simple-cubic LGF operator is irreducible and non-Liouvillian yet closed-form in  $K$ . Indeed, as the next section shows, the hyperkagome operator *is* orthogonally equivalent to a symmetric square, and Section 6 proves the resulting closed form outright—it is modular, through eta quotients and genus-zero modular functions, which are themselves non-Liouvillian; the non-Liouvillian result excludes only the algebraic and elementary cases and is fully consistent with modularity.

## 5. An invariant quadratic form: the orthogonal Galois group

The symmetric square  $\mathrm{Sym}^2(M)$ —the order-six operator annihilating the pairwise products  $y_i y_j$  of solutions of  $M$ —possesses an explicit *rational* solution,<sup>1</sup>

$$(6) \quad R(t) = \frac{(15t^2 + 17t - 8)^2}{t^2 (t - 1)^2 (4t - 1)(5t - 1)(9t - 1)}.$$

Being the solution of a linear equation,  $R$  is fixed only up to an overall constant  $\lambda$ , and so is the Gram matrix it determines: expanding  $\lambda R = \sum_{i \leq j} c_{ij} y_i y_j$  in a fundamental system  $y_1, y_2, y_3$  of  $M$  gives a symmetric  $3 \times 3$  matrix  $C = (c_{ij})$ . We record it for the normalization  $\lambda = -1/272$ , which is the one carried by the accompanying certificate. In the basis fixed by the initial conditions  $y_i^{(j)}(t_0) = \delta_{ij}$  at the ordinary point  $t_0 = \frac{1}{2}$ ,

$$(7) \quad (c_{11}, c_{12}, c_{13}, c_{22}, c_{23}, c_{33}) = \left( -\frac{17}{84}, -\frac{460}{441}, \frac{44927}{3087}, -\frac{1925200}{157437}, \frac{83066560}{1102059}, -\frac{712021969}{7714413} \right),$$

$$\det C = \frac{3437476900}{7714413} \neq 0 \text{ (rank 3)}.$$

Equation (6) is verified as an exact identity: the six constants are fixed by six series coefficients and then satisfy 194 further exact relations at each of two independent ordinary base points ( $t_0 = \frac{1}{2}$  and  $t_0 = -\frac{1}{3}$ ). The agreement is a proof, not merely a margin, by a Fuchs-relation budget: were  $F = \lambda R - \sum c_{ij} y_i y_j$  not identically zero, the span of the six products and  $R$  would be seven-dimensional, with a Fuchsian minimal operator of order seven whose

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<sup>1</sup>The homomorphism of  $M$  to its adjoint, the rational solution  $R$  of  $\mathrm{Sym}^2(M)$ , and the explicit order-two intertwiner realizing it are certified in exact arithmetic in the accompanying repository.

local exponents are bounded below by the certified data—the pairwise sums of the exponents of (4) at each singular point, the pole and zero orders of  $R$ , and *distinct* non-negative integers at every apparent point. The Fuchs relation then caps the vanishing order of any nonzero element of that span at an  $M$ -ordinary point at 109, while  $F$  is verified to vanish to order 200; hence  $F \equiv 0$  identically. Independently and at the operator level, we exhibit an order-two differential intertwiner  $\iota \in \text{Hom}(\text{adjoint}(M), M)$  and verify the exact identity  $\text{rightrem}(M \iota, \text{adjoint}(M)) = 0$  over  $\mathbb{Q}(t)$ , so  $M$  is homomorphic—in fact isomorphic, since both are irreducible of the same rank—to its adjoint. Four consequences follow.

(1) *The monodromy lies in an orthogonal group.* Because  $R$  is single-valued (rational) and  $R = \sum c_{ij} y_i y_j$  with the six products linearly independent (the leading  $6 \times 6$  coefficient system is nonsingular), every monodromy matrix  $\gamma$  obeys  $\gamma^\top C \gamma = C$ : the solution space carries a monodromy-invariant symmetric bilinear form.  $M$  is Fuchsian, so by the density theorem the differential Galois group  $G_M$  is the Zariski closure of the monodromy group, whence  $G_M \subseteq \text{O}(C) \cong \text{O}(3, \mathbb{C})$ .

(2) *The form is nondegenerate.* By irreducibility, the kernel of  $C$ —an invariant subspace—is 0, so  $C$  is nondegenerate; by Schur's lemma the invariant form is unique up to scale, and in odd dimension it is symmetric rather than symplectic. We confirm  $\det C \neq 0$  directly, at both base points.

(3) *The identity component is  $\text{SO}(3, \mathbb{C})$ .* The genuine logarithm at  $t = 0$  makes a monodromy element non-semisimple, so  $G_M$  is infinite. Suppose, for contradiction, that  $G_M^\circ$  were solvable. As a normal subgroup of  $G_M$ ,  $G_M^\circ$  acts semisimply on the solution space  $V$  by Clifford's theorem (since  $V$  is  $G_M$ -irreducible); by the Lie–Kolchin theorem [32] the irreducible representations of a connected solvable group are one-dimensional, so  $G_M^\circ$  acts diagonally in some basis and  $G_M$  permutes the resulting isotypic components. If the three characters are distinct,  $G_M$  is monomial (imprimitive)—but every monomial matrix is semisimple (if its permutation part has order  $k$ , its  $k$ -th power is diagonal and invertible, and an invertible matrix with a diagonalizable power is diagonalizable in characteristic zero), contradicting the logarithm. A (2+1) pattern of characters cannot be permuted nontrivially, which would make  $V$  reducible—again a contradiction. A single character makes  $G_M^\circ$  scalar, hence, inside  $\text{O}(3, \mathbb{C})$ , finite, forcing  $G_M$  finite and every element semisimple—once more contradicting the logarithm. Hence  $G_M^\circ$  is non-solvable, and the only non-solvable connected subgroup of  $\text{O}(3, \mathbb{C})$  is  $\text{SO}(3, \mathbb{C})$ ; therefore  $G_M^\circ = \text{SO}(3, \mathbb{C})$ .

(4) *The determinant character:  $G_M = \text{O}(3, \mathbb{C})$  in full.* The Wronskian  $\text{Wr}$  of  $M$  satisfies  $\text{Wr}'/\text{Wr} = -c_2/c_3$ , whose residue at each singular point

equals the local exponent sum there minus 3. This is half-integral exactly at  $t = \frac{1}{9}, \frac{1}{5}, \frac{1}{4}$  and  $\infty$  (for instance at  $\frac{1}{9}$ ,  $(-\frac{1}{2} + 0 + 1) - 3 = -\frac{5}{2}$ ) and integral at  $t = 0, 1$  and on  $p_7$ ; hence  $\det(\text{monodromy}) = -1$  at those four points and  $+1$  elsewhere. Thus  $G_M$  is *not* contained in  $\text{SO}(3, \mathbb{C})$ , i.e.  $G_M = \text{O}(3, \mathbb{C})$  in full, and the determinant character is the quadratic character of the genus-one curve

$$(8) \quad v^2 = (1 - 4t)(1 - 5t)(1 - 9t),$$

ramified at exactly  $\{\frac{1}{9}, \frac{1}{5}, \frac{1}{4}, \infty\}$ —an even number of points, as a double cover requires, and the same four points at which the pole orders of  $R$  in (6) are odd.

Since  $\text{SO}(3, \mathbb{C}) \cong \text{PSL}(2, \mathbb{C})$  through the symmetric square of the standard two-dimensional representation,  $G_M^\circ = \text{SO}(3, \mathbb{C})$  means that  $M$  is *projectively equivalent* to  $\text{Sym}^2(V_2)$  for a second-order operator  $V_2$  (defined over  $\mathbb{Q}(t, v)$  with  $v$  as in (8), or over  $\mathbb{Q}(t)$  after the corresponding quadratic twist). The equivalence is carried by an explicit intertwiner, but whether that intertwiner is *rational* depends on how  $V_2$  is normalized, and the point is worth making plainly because the search that a reader would run first returns nothing. Against  $\text{Sym}^2(V_2)$  with  $V_2$  in the projective normal form  $w'' + Q_V w = 0$  of (9) there is no homomorphism over  $\mathbb{Q}(t)$  in either direction; after the single conjugation  $M_v := v \circ M \circ v^{-1}$ —which again has coefficients in  $\mathbb{Q}(t)$ , because  $v'/v$  is rational—the intertwiner is the *order-one* operator  $T = \rho_0 + \rho_1 \text{d}/\text{d}t$  of (15) below; Remark 5.1 says why, and why the obstruction disappears one level up. The equivalence shifts individual local exponents by *different* integers and so destroys the arithmetic progressions, which is exactly why  $M$  is not a literal symmetric square (nor a multiplier or pullback of one) yet is orthogonally, projectively, one. The same structural phenomenon is known in the differential algebra of lattice models: Assis, van Hoeij and Maillard [4] encounter an order-three operator  $N_3$  whose symmetric square has a rational solution,<sup>2</sup> hence is homomorphic to a symmetric square  $\text{Sym}^2(V_2)$ ; they recover  $V_2$  by a conic construction, and solve it as a  ${}_2F_1$  with an algebraic (modular) pullback. For the hyperkagome that conic construction has now been carried out in the accompanying analysis: the transported invariant form admits an explicit rational conic point, yielding an *explicit* second-order operator  $V_2$  over

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<sup>2</sup>Section 4.2 of [4] attributes that rational solution to the *exterior* square of  $N_3$ ; this is a misprint for the *symmetric* square, as the remainder of the same sentence—which concludes that  $N_3$  is homomorphic to a symmetric square—makes clear. For an irreducible order-three operator the exterior square is isomorphic to the dual twisted by the Wronskian and can carry no rational solution, whereas a rational solution of the symmetric square is precisely the invariant quadratic form used here.

$\mathbb{Q}(t)$  (the quadratic twist enters only through the pullback), taken here in the form  $d^2/dt^2 + a_1 d/dt + a_0$  of `numerics/V2_data.json` rather than in the projective normal form (9), a distinction that matters for the intertwiner exactly as noted above; it is verified in exact arithmetic through the bridge identity  $f_0^2 = P(y_0)$  with  $V_2(f_0) = 0$  on every Laurent coefficient through order  $t^{107}$ , where  $P$  is an explicit order-two intertwiner and  $y_0 = \Phi'/2$  the analytic solution. The operator  $V_2$ , the rational conic point, and this verification are public: `numerics/V2_data.json` and `numerics/certify_bridge.py` in the repository, the latter validating each of its series primitives on operators of known structure, and failing on a deliberately perturbed conic point, before the certified run. The pullback question is settled, in its sharpest form, in Section 6:  $V_2$  is the uniformizing operator of the modular curve  $X(\Gamma_0(30)^+)$ , its mirror map at the maximally unipotent point  $t = 0$  [29] (see below) is the level-30 eta-quotient parametrization (11), and the sought closed form is modular—an Assis–van Hoeij–Maillard-style  ${}_2F_1$  with *algebraic* pullback is now explicit:  $1728/j$  is algebraic of degree eight over  $\mathbb{Q}(t)$  and solvable in radicals (Proposition 6.2)—though the modular parametrization itself remains the closed form in its natural coordinates.<sup>3</sup>

*Remark 5.1* (Where the rational intertwiner hides). The direct search returns nothing, and the reason is structural rather than computational. With  $V_2$  in the projective normal form  $w'' + Q_V w = 0$  of (9), the operator  $\text{Sym}^2(V_2) = d^3/dt^3 + 4Q_V d/dt + 2Q'_V$  is unimodular, whereas the Wronskian logarithmic derivative of  $M$  has residue  $-5/2$  at each of  $t = \frac{1}{4}, \frac{1}{5}, \frac{1}{9}$ . No residue of  $f'/f$  can fail to be an integer, so  $\Lambda^3(M)$  is non-trivial, and two irreducible operators with non-isomorphic determinants are non-isomorphic (for  $\text{Sym}^2(V_2)$  the irreducibility is Lemma 7.2 below): in Maple, `Homomorphisms(symmetric_power(V2,2),M)` is empty in both directions of necessity. What removes the obstruction is that the determinant character of  $M$  is the quadratic character  $\chi_v$  of (8): conjugating by  $v$  multiplies it by  $\chi_v^3 = \chi_v$ , leaving  $\chi_v^2 = 1$ . Concretely the logarithmic derivative shifts by  $3v'/v$ , and the certificate exhibits an explicit  $f \in \mathbb{Q}(t)^*$  with  $f'/f$  equal to the result. The intertwiner is then found by `Homomorphisms(symmetric_power(V2,2),mult(v,M,1/v))` with  $v = (1 - 4t)^{1/2}(1 - 5t)^{1/2}(1 - 9t)^{1/2}$ , and is  $T$ . The same parity removes the

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<sup>3</sup>The base  ${}_2F_1$  is a choice, and it is not a neutral one. Taking  ${}_2F_1(\frac{1}{8}, \frac{3}{8}; 1; \cdot)$  in place of  ${}_2F_1(\frac{1}{12}, \frac{5}{12}; 1; \cdot)$  gives a markedly simpler exact expression: the pullback then satisfies an equation of bidegree  $(4, 24)$  rather than  $(8, 72)$ , over an index-two subfield of (13) reached by two square roots instead of three. See the footnote to Proposition 6.2.

obstruction one level up, and on both sides:  $\det \text{Sym}^2 = (\det)^4$  trivializes the determinant of  $\text{Sym}^2(M)$  because  $\chi_v^4 = 1$ , and  $\text{Sym}^4$  of a rank-two operator with trivial determinant again has trivial determinant ( $\det \text{Sym}^4 = (\det)^{10}$ ). That is only the removal of an obstruction, not a construction; the homomorphism between  $\text{Sym}^2(M)$  and  $\text{Sym}^4(V_2)$  is easy to check in Maple and is not certified here, whereas  $M$  and  $\text{Sym}^2(V_2)$  are provably not homomorphic. There is no tension with the rational order-two  $P$  of the bridge identity  $f_0^2 = P(y_0)$  above: that  $P$  intertwines  $M$  with the symmetric square of  $V_2$  in the *non-unimodular* normalization  $d^2/dt^2 + a_1 d/dt + a_0$ , whose determinant character differs from that of  $M$  by integer residues only, so there the obstruction is absent. That  $P$  is an intertwiner in the module sense, and not merely the single-solution statement  $f_0^2 = P(y_0)$ , is itself an exact operator identity: the right remainder of  $\text{Sym}^2(V_2) \circ P$  on division by  $M$  is zero, whereas the same  $P$  against the projective normal form leaves a nonzero remainder. It is precisely the passage to the projective normal form that reintroduces the obstruction. The computational content of this remark—the residues, the determinant non-isomorphy (which yields the non-existence of the rational intertwiner given the two irreducibility inputs already cited), the trivialization of the character, the order-one  $T$ , and this last consistency—is certified in exact arithmetic, with negative controls, in `numerics/certify_intertwiner.py`, which builds  $M_v$  by expanding  $(d/dt - v'/v)^i$  and so never leaves  $\mathbb{Q}(t)$ ; the Maple invocation is quoted as a convenience and is not itself certified.

*Maximal unipotent monodromy at  $t = 0$  ( $n = 2$ ).* The three exponents at  $t = 0$ ,  $\{-1, -1, 0\}$  (indicial polynomial  $-64\rho(\rho+1)^2$ ), are all integers, so the local monodromy is unipotent. Exactly one of the three local solutions is log-free (single-valued Laurent); since the number of log-free solutions equals the number of Jordan blocks of a unipotent, the monodromy is a *single*  $3 \times 3$  block: the point is *maximally unipotent* (MUM), with maximal log power  $n = 2$ . It is MUM in the monodromy sense with exponents  $\{-1, -1, 0\}$  rather than  $\{0, 0, 0\}$ : an integer recombination of the Frobenius solutions brings the basis to the canonical MUM normal form  $\{y_0, y_0 \log t + f_1, \frac{1}{2}y_0 \log^2 t + f_1 \log t + f_2\}$ , with the *same* series  $f_1$  multiplying the logarithm in the third solution as appears in the second,<sup>4</sup> where  $y_0 = \Phi'/2$  is the holomorphic solution and  $f_1, f_2$  carry  $1/t$  Laurent heads. This settles a point that was initially unclear:  $n = 2$ , not 1. The even value  $n = 2$  is consistent with the orthogonal (rather

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<sup>4</sup>The recombination ( $\tilde{y}_2 = y_2 - \frac{13}{30}y_1$  in the natural rescaling) is verified exactly, term by term and through the identity  $M(y_0 \log t + f_1) = 0$  order by order, in `verify_mum_normalform.py` in the repository.

than symplectic) case of the conjectured parity correspondence of Hassani, Maillard and Zenine [17], in which the parity of the maximal log power tracks the orthogonal/symplectic type of the differential Galois group. In the energy variable  $t = (E - 1)^{-2}$  the point  $t = 0$  is  $E = \infty$  (equivalently  $z = \infty$ ), so this is the local expansion at infinity, and the  $1/t$  Laurent heads of the logarithmic partners are a feature of that point rather than a pathology.

## 6. The modular parametrization: $V_2$ uniformizes $X(\Gamma_0(30)^+)$

We now prove the statement toward which Section 5 points, and which the lattice-side reduction of Section 9 below independently anchors: the second-order operator  $V_2$  is the *uniformizing differential equation* of a classical modular curve, and the variable  $t$  generates its genus-zero function field.<sup>5</sup> Throughout,  $\tau$  lies in the upper half-plane,  $q = e^{2\pi i\tau}$ ,  $\eta(\tau) = q^{1/24} \prod_{n \geq 1} (1 - q^n)$ , and  $\Gamma_0(30)^+$  denotes the extension of  $\Gamma_0(30)$  by its seven Atkin–Lehner involutions  $w_e$ ,  $e \in \{2, 3, 5, 6, 10, 15, 30\}$ , of which  $w_{30}$  is the *Fricke involution* [5, 11]; we henceforth write  $v$  for the twist variable of (8), reserving  $u$  for the modular function below. The projective normal form of the explicit  $V_2$  of Section 5—the operator  $w'' + Q_V(t)w = 0$  obtained from  $V_2 = \partial_t^2 + a_1\partial_t + a_0$  via  $Q_V = a_0 - \frac{1}{4}a_1^2 - \frac{1}{2}a_1'$ , invariant under every function-multiplier gauge and hence a projective invariant of  $M$  itself—is

$$(9) \quad \begin{aligned} Q_V(t) &= \frac{N(t)}{4t^2(t-1)^2(4t-1)^2(5t-1)^2(9t-1)^2}, \\ N(t) &= 24300t^8 - 58860t^7 + 73437t^6 - 44294t^5 \\ &\quad + 15111t^4 - 3160t^3 + 407t^2 - 30t + 1. \end{aligned}$$

The double poles of  $Q_V$  sit *exactly* at the six points  $t = 0, \frac{1}{9}, \frac{1}{5}, \frac{1}{4}, 1, \infty$ : the seven apparent singularities on  $p_7$ , and every trace of the order-two intertwiner, cancel identically in the normal form. The Laurent head of  $Q_V$  at each of  $t = \frac{1}{9}, \frac{1}{5}, \frac{1}{4}, 1, \infty$  is  $\frac{3}{16}$  (local exponent difference  $\frac{1}{2}$ ), and at  $t = 0$  it is

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<sup>5</sup>Terminology: in the Conway–Norton normalization [11] a degree-one generator of the genus-zero function field of  $X(\Gamma)$  is called a *Hauptmodul* of  $\Gamma$ , and in that sense  $t$  is one for  $\Gamma_0(30)^+$ . We reserve the word here for the pullback  $1728/j$  that arises as the argument  $H$  of  ${}_2F_1(\frac{1}{12}, \frac{5}{12}; 1; H)$  in an Assis–van Hoeij–Maillard-style representation [4], and refer to the explicit map (10)–(11) as a *Weber-like function parametrization*.



$\frac{1}{4}$  (exponent difference 0: the MUM point). Define the level-30 eta quotient

$$(10) \quad u(\tau) = \prod_{d|30} \eta(d\tau)^{r_d} = \left[ \frac{\eta(\tau) \eta(6\tau) \eta(10\tau) \eta(15\tau)}{\eta(2\tau) \eta(3\tau) \eta(5\tau) \eta(30\tau)} \right]^3,$$

i.e.  $(r_d) = (3, -3, -3, -3, 3, 3, 3, -3)$  on  $d = (1, 2, 3, 5, 6, 10, 15, 30)$ , and set

$$(11) \quad t = \frac{u}{u^2 + 7u + 1}, \quad \text{equivalently} \quad \frac{1}{t} = u + 7 + \frac{1}{u},$$

so that  $t = q - 4q^2 + 12q^3 - 34q^4 + 90q^5 - \dots$ . We refer to (10)–(11) as a *Weber-like function parametrization* of  $t$ .

The generator so obtained is a classical object rather than a new one, and we record the identification explicitly. The eta quotient (10) is the McKay–Thompson series of class 30A for the Monster, and  $u + 3 + 1/u$  is that of class 30B [14]; in our normalization, which places the cusp at  $t = 0$ , one has  $1/t = T_{30B} + 4$ , the two conventions differing only in the  $q^0$  coefficient. What is at stake here is therefore not the  $q$ -series but the *proof* that this particular modular function is the one the lattice Green’s function selects.

**Theorem 6.1.**  *$1/t$  generates the genus-zero function field of  $X(\Gamma_0(30)^+)$  (so  $t$  is a degree-one generator vanishing at the unique cusp), and the Schwarzian identity*

$$(12) \quad \{\tau, t\} = 2Q_V(t), \quad \{\tau, t\} \equiv \frac{\tau'''}{\tau'} - \frac{3}{2} \left( \frac{\tau''}{\tau'} \right)^2, \quad ' = \frac{d}{dt},$$

*holds exactly.*

The identity (12) is of the type studied in [1, 2]. The Schwarzian route from an order-three operator with orthogonal Galois group, through the symmetric square of an underlying order-two operator, to a modular correspondence is developed in [1, 2]; (12) is an instance in which the correspondence is pinned to a named group and proved rather than exhibited.

We note that the right-hand side of (12) is not new as a  $q$ -series: a Schwarzian  $Q$ -value for every genus-zero Conway–Norton class, including 30B, is tabulated in [25], and that entry is exactly our  $Q_V$  under the change of generator  $t = z/(1 + 4z)$  induced by  $1/t = T_{30B} + 4$ . That table is obtained by fitting an ansatz to a bounded  $q$ -expansion, with no degree bound and no proof of termination. The contribution of (12) is the a priori pole-degree bound established in the proof of Theorem 6.1, which converts such a match into an identity; in particular it retro-proves the tabulated 30B entry. Equivalently:



the ratio of two solutions of  $w'' + Q_V w = 0$  inverts the parametrization  $t(\tau)$ ,  $V_2$  is the uniformizing operator of the orbifold  $X(\Gamma_0(30)^+)$ , and the mirror map of  $V_2$  at its MUM point  $t = 0$  is precisely (11), with the modular  $q$  as nome. (The identification was found by matching the mirror map against (10)–(11) to all 80 computed orders; the proof below shows the match is forced.)

*Proof. Modular part.* The exponent vector  $(r_d)$  satisfies Ligozat's criterion [26]:  $\sum_d d r_d = -24 \equiv 0$  and  $\sum_d (30/d) r_d = 24 \equiv 0 \pmod{24}$ , the weight  $\frac{1}{2} \sum_d r_d$  vanishes, and  $\prod_d d^{r_d} = 1$  is a rational square; hence  $u$  is a modular function on  $\Gamma_0(30)$ . Ligozat's cusp-order formula gives its divisor on  $X_0(30)$  (cusps labeled by their denominators  $d \mid 30$ ): simple zeros at the cusps  $\{1, 6, 10, 15\}$  and simple poles at  $\{2, 3, 5, 30\}$ , so  $u$  has degree four. For the Fricke involution, the exact antisymmetry  $r_{30/d} = -r_d$  forces  $w_{30}(u) = 1/u$  structurally (the eta multipliers cancel because  $\sum_d r_d = 0$  and  $\prod_d (30/d)^{r_d} = 1$ ); comparing divisors and leading Fourier coefficients—two modular functions with equal divisors and equal leading coefficient are equal—shows likewise that  $w_2, w_3, w_5$  each send  $u \mapsto 1/u$ , whence the complementary products  $w_6, w_{10}, w_{15}$  fix  $u$ . Since  $\{1, w_6, w_{10}, w_{15}\}$  is an index-two subgroup of the Atkin–Lehner group  $\mathcal{W} \cong (\mathbb{Z}/2)^3$ ,  $u$  descends to a function of degree  $4/4 = 1$  on  $X(\Gamma_0(30)+6, 10, 15)$ : it generates the function field there, and that curve has genus zero. Consequently  $1/t = u + 1/u + 7$  is invariant under all of  $\Gamma_0(30)^+$  and descends to degree one on the quotient: it generates the function field of  $X(\Gamma_0(30)^+)$ , and that curve has genus zero. It has a single cusp:  $w_e$  maps the cusp of denominator  $d$  to that of denominator  $de/\gcd(d, e)^2$ , a simply transitive action of  $\mathcal{W}$  on the eight cusps of  $X_0(30)$ . The normalization (11) puts the cusp at  $t = 0$ , with  $t = q + O(q^2)$  (cusp width one).

*Uniformization part.* First, the signature of  $\Gamma_0(30)^+$  is forced.  $\Gamma_0(30)$  is torsion-free: for  $30 = 2 \cdot 3 \cdot 5$  the elliptic-point counts are  $\nu_2 = \prod_{p \mid 30} (1 + (\frac{-1}{p})) = 0$  (the factor at  $p = 3$  vanishes) and  $\nu_3 = \prod_{p \mid 30} (1 + (\frac{-3}{p})) = 0$  (the factor at  $p = 2$  vanishes) [12]. Any elliptic  $\gamma \in \Gamma_0(30)^+$  has  $\gamma^2 \in \Gamma_0(30)$ , and  $\gamma^2$  is elliptic or the identity; torsion-freeness forces  $\gamma^2 = 1$ , so every elliptic point of  $\Gamma_0(30)^+$  has order two. The orbifold Euler characteristic is  $\chi = \frac{1}{8} \chi(\Gamma_0(30)) = -\frac{1}{8} \cdot \frac{72}{6} = -\frac{3}{2}$  (the index is  $[\mathrm{PSL}_2(\mathbb{Z}) : \Gamma_0(30)] = 72$ ), i.e. hyperbolic covolume  $3\pi$ —*nine* times that of  $\mathrm{PSL}_2(\mathbb{Z})$ , a covolume ratio and not a subgroup index, since  $\Gamma_0(30)^+$  is not a subgroup of  $\mathrm{PSL}_2(\mathbb{Z})$ . With genus zero and one cusp, Gauss–Bonnet  $2 - 1 - \sum_i (1 - e_i^{-1}) = -\frac{3}{2}$  gives exactly *five* elliptic points, all of order two: the signature  $(0; 2, 2, 2, 2, 2; 1 \text{ cusp})$  is derived, not assumed. Second, the identity (12) follows from a degree count. Because  $t$  is a degree-one function on the genus-zero quotient of a finite-covolume

Fuchsian group,  $\{\tau, t\}$  is a *rational* function of  $t$ , with double poles precisely at the images of the elliptic points and of the cusp, and with universal Laurent heads:  $\frac{1}{2}(1-e^{-2})$  at an order- $e$  elliptic image and  $\frac{1}{2}$  at a cusp image—so  $\frac{1}{2}\{\tau, t\}$  carries heads  $\frac{3}{16}$  and  $\frac{1}{4}$ , matching (9). Consider  $D(t) = \{\tau, t\} - 2Q_V(t)$ , a rational function. Its poles are confined to the five (a priori unlocated) elliptic images, the cusp image  $t = 0$ , the remaining finite poles  $\frac{1}{9}, \frac{1}{5}, \frac{1}{4}, 1$  of  $Q_V$ , and  $t = \infty$ , each of order at most two; at  $t = 0$  the heads  $\frac{1}{2t^2}$  cancel exactly, leaving at most a simple pole. The pole divisor of  $D$  on  $\mathbb{P}^1$  therefore has degree at most  $2(5 + 4 + 1) + 1 = 21$ . On the other hand we verify, in exact rational arithmetic, that the  $q$ -expansion of  $D(t(q))$ —computable term by term from (10)–(11)—vanishes through order  $q^{80}$ ; since  $t(q) = q + O(q^2)$  is a formal immersion,  $D$  vanishes at  $t = 0$  to order at least 80. A nonzero rational function has equal degrees of zeros and poles, and  $80 > 21$ ; hence  $D \equiv 0$ .  $\square$

The proof is exact and non-numerical throughout: the modular part is finite eta-quotient combinatorics, and the uniformization part is a finite exact series computation closed off by the degree bound. As a corollary the double poles of  $\{\tau, t\} = 2Q_V$  are exactly  $t = 0, \frac{1}{9}, \frac{1}{5}, \frac{1}{4}, 1, \infty$ , so the five order-two points of  $\Gamma_0(30)^+$  project precisely to  $\{\frac{1}{9}, \frac{1}{5}, \frac{1}{4}, 1, \infty\}$  and the cusp to  $t = 0$ : the singular locus of  $V_2$  (equivalently of  $M$ , up to the apparent  $p_7$ ) is the orbifold locus of  $X(\Gamma_0(30)^+)$ , with no elliptic point hiding elsewhere. In  $u$ -coordinates the dictionary is explicit:  $t = \frac{1}{9} \leftrightarrow u = 1$  and  $t = \frac{1}{5} \leftrightarrow u = -1$  (the branch points of  $u \mapsto u + 1/u$ ), while  $t = \frac{1}{4}, 1, \infty$  correspond to the root pairs of  $u^2 + 3u + 1, u^2 + 6u + 1, u^2 + 7u + 1$ , and the cusp  $t = 0$  to  $u \in \{0, \infty\}$ .

Three consequences deserve separate statement. (i) *The monodromy is arithmetic.* The projective monodromy group of  $V_2$ —hence of  $M$ , by the projective equivalence of Section 5—is the deck group of the uniformization: *exactly* the discrete arithmetic lattice  $\Gamma_0(30)^+ \subset \mathrm{PSL}_2(\mathbb{R})$ , of covolume  $3\pi$ . (ii) *The twist is itself modular.* The determinant character of Section 5 is a quadratic character of  $\Gamma_0(30)^+$  trivial on  $\Gamma_0(30)$ ; its kernel is the index-two subgroup  $\Gamma_0(30)+2, 3, 6$ , and the associated double cover of the  $t$ -line, branched at the four  $\det = -1$  points  $\{\frac{1}{9}, \frac{1}{5}, \frac{1}{4}, \infty\}$ , has genus one by Riemann–Hurwitz—it is precisely the twist curve  $v^2 = (1 - 4t)(1 - 5t)(1 - 9t)$  of (8), now identified as  $X(\Gamma_0(30)+2, 3, 6)$ . In particular the analytic period  $y_0 = \Phi'/2$ , pulled back by (11), is a weight-two object carrying this quadratic twist character, in the manner familiar from the modular interpretation of the Apéry numbers [6]—though, as Section 7 shows, a *depth-one quasimodular* one [35] rather than a modular form; its explicit closed form is derived there. (iii) *The Varma–Monien question is resolved.* Varma and Monien judged it “highly suggestive that a closed form expression can be obtained” for their

threefold integral [33]; the theorem proves this in the strongest structural sense, with the closed form modular rather than elliptic: the hyperkagome LGF lives at level  $30 = 2 \cdot 3 \cdot 5$ . To our knowledge every lattice Green's function (and sunrise-integral period) previously realized modularly occurs at a level with at most two distinct primes—levels 2, 3, 4, 6 dominate the classical cubic-lattice and Bessel-moment literature [10, 7]—so the hyperkagome appears to be the first lattice Green's function realized at a level with three distinct prime factors. The Assis–van Hoeij–Maillard-style  ${}_2F_1$  representation [4] is now explicit rather than merely in-principle, and its algebraic pullback turns out to be solvable in radicals.

**Proposition 6.2** (The pullback in radicals). *The pullback  $H = 1728/j$ , the argument of the  ${}_2F_1(\frac{1}{12}, \frac{5}{12}; 1; \cdot)$  representation, is algebraic of degree exactly eight over  $\mathbb{Q}(t)$ : its primitive minimal polynomial  $P_H(H, t)$  has bidegree  $(8, 72)$ , and the splitting field of  $P_H$  is the multiquadratic extension*

$$(13) \quad \mathbb{Q}(t) \left( \sqrt{(1-t)(1-9t)}, \sqrt{(1-t)(1-5t)}, \sqrt{1-4t} \right),$$

*which is the full function field of  $X_0(30)$ . Consequently  $\text{Gal}(P_H/\mathbb{Q}(t)) \cong (\mathbb{Z}/2)^3$  is the Atkin–Lehner group  $\mathcal{W}$ , and the  ${}_2F_1$  pullback is solvable in radicals.*

The product of the three radicands is  $((1-t)v)^2$ : the determinant-character twist  $v$  of (8) is, up to sign and the rational factor  $1-t$ , the product of the three quadratic layers of the radical tower.<sup>6</sup>

*Proof.* The coefficients of  $P_H$ , an explicit radical expression for  $H$  due to M. van Hoeij (private communication, July 2026), and the machine certificate are public (`numerics/pullback_data.json`, `numerics/certify_pullback.py`). The certificate proves four exact statements. (i)  $P_H(H, t) = 0$  for the modular  $H$ : the identity is verified on the  $q$ -expansions through  $q^{1200}$ , and this is a proof rather than a series check because  $H$  lies in the function field of  $X_0(30)$ , of degree  $|\mathcal{W}| = 8$  over  $\mathbb{Q}(t)$  (quotient covolume ratio  $24\pi/3\pi$ ), with pole divisor of degree 72:  $j$  has degree  $[\text{PSL}_2(\mathbb{Z}) : \Gamma_0(30)] = 72$ , and  $\nu_3(30) = 0$  means that no point above  $j = 0$  retains an order-three stabilizer, so all  $72/3 = 24$  of them are ramified of index three and  $H = 1728/j$  has

<sup>6</sup>For the equivalent base  ${}_2F_1(\frac{1}{8}, \frac{3}{8}; 1; \cdot)$  the pullback satisfies a smaller equation, of bidegree  $(4, 24)$ , whose two radicands  $(1-t)(1-5t)$  and  $(1-t)(1-4t)(1-9t)$  generate an index-two subfield of (13); its analytic branch  $\tilde{H} = 256t - 25600t^2 + \dots$  is verified against the certified  $Q_V$  through the Schwarzian pullback identity to order  $t^{96}$ .

24 triple poles. Hence a nonzero polynomial of bidegree at most  $(8, 72)$  evaluated on  $H$  has polar degree at most  $8 \cdot 72 + 72 \cdot 8 = 1152$ , and since a nonzero function on a compact curve has as many zeros as poles, its order of vanishing at the cusp—where  $q$  is a uniformizer,  $\Gamma_0(30)$  having width one at  $\infty$ —is at most  $1152 < 1200$ . (ii) The radical expression is an exact root of  $P_H$ : a polynomial identity in the multiquadratic algebra, in exact integer arithmetic, no series involved. (iii) The seven subset products of the radicands are non-squares, so (13) has degree eight over  $\mathbb{Q}(t)$ . (iv) The eight sign-conjugates of the radical expression are pairwise distinct, and the branch with signs  $(-, -, -)$  matches the  $t$ -expansion of the modular  $H$ ; hence  $H$  has degree exactly eight,  $P_H$  is irreducible and is the minimal polynomial, and  $\mathbb{Q}(t)(H)$  is all of (13). That field is  $\mathbb{Q}(X_0(30))$ : by Theorem 6.1  $\mathbb{Q}(t)$  is the fixed field of  $\mathcal{W}$  acting on  $\mathbb{Q}(X_0(30))$ , so  $[\mathbb{Q}(X_0(30)) : \mathbb{Q}(t)] = |\mathcal{W}| = 8$  by Artin's theorem, while  $\mathbb{Q}(t)(H) \subseteq \mathbb{Q}(X_0(30))$  because  $H = 1728/j$  is a modular function for  $\Gamma_0(30)$ ; equal degrees force equality, and  $\text{Gal} = \mathcal{W}$ . As a consistency check, Riemann–Hurwitz for the  $(\mathbb{Z}/2)^3$ -cover (13) of the  $t$ -line, branched over the five order-two orbifold points— $\{1, \frac{1}{9}\}$ ,  $\{1, \frac{1}{5}\}$  and  $\{\frac{1}{4}, \infty\}$  through the three quadratic layers—gives four double points over each, hence  $2g - 2 = 8 \cdot (-2) + 5 \cdot 4$ , i.e.  $g = 3$ , the classical genus of  $X_0(30)$ . The ramification index is two even at  $t = 1$ , where two of the radicands vanish simultaneously: their product  $(1 - t)^2(1 - 5t)(1 - 9t)$  has *even* valuation there, so of the seven nontrivial characters exactly four are ramified and the unramified ones form a subgroup of order four; the inertia group is generated by the simultaneous sign flip and has order two, not four.  $\square$

The representation (10)–(11) remains the closed form in its most natural coordinates; the proposition shows what replaces rationality of the pullback: an abelian  $(\mathbb{Z}/2)^3$  radical tower whose quadratic layers are cut out by the Atkin–Lehner involutions, with the determinant-character twist  $v$  appearing as their product.

## 7. The weight-two period in closed form

The weight-two period  $y_0 = \Phi'/2$  itself admits an explicit closed form in the modular data of Section 6. Write  $W = q dt/dq$  for the nome derivative of the modular function  $t$  of (11) (so that  $W$  is a constant multiple of  $w_1^2$ , with  $w_1 = (dt/d\tau)^{1/2}$  a solution of the projective normal form  $w'' + Q_V w = 0$  of (9)), put  $W' = dW/dt$ , and let  $v$  be the twist unit of (8) in the branch  $v(0) = +1$ .

**Theorem 7.1.** *With this notation, the weight-two period of the hyperkagome lattice Green's function is*

$$(14) \quad y_0 = \frac{\rho_0(t) W + \rho_1(t) W'}{v},$$

$$(15) \quad \begin{aligned} \rho_1(t) &= \frac{15t^2 + 17t - 8}{30t(t-1)}, \\ \rho_0(t) &= -\frac{4050t^6 + 2445t^5 - 11436t^4 + 8000t^3 - 2130t^2 + 231t - 8}{30t^2(t-1)^2(4t-1)(5t-1)(9t-1)}. \end{aligned}$$

Equivalently, cleared of all denominators,

$$(16) \quad B(t) y_0 v W = A_0(t) W^2 + A_1(t) q \frac{dW}{dq},$$

with  $B = -30t^2(t-1)^2(4t-1)(5t-1)(9t-1)$ ,  $A_0 = 4050t^6 + 2445t^5 - 11436t^4 + 8000t^3 - 2130t^2 + 231t - 8$ , and  $A_1 = -t(t-1)(4t-1)(5t-1)(9t-1)(15t^2 + 17t - 8)$ , using  $q dW/dq = WW'$ . Every ingredient is an explicit level-30 modular or quasimodular object: with  $u$  the eta quotient (10),  $E_2(\tau) = 1 - 24 \sum_{n \geq 1} \sigma_1(n) q^n$ , and the Eisenstein-series form of the  $\eta$ -logarithmic derivative

$$(17) \quad \begin{aligned} \ell = q \frac{d \log u}{dq} &= \frac{1}{8} [E_2(\tau) + 6E_2(6\tau) + 10E_2(10\tau) + 15E_2(15\tau) \\ &\quad - 2E_2(2\tau) - 3E_2(3\tau) - 5E_2(5\tau) - 30E_2(30\tau)], \end{aligned}$$

one has  $W = (1 - u^2) u \ell / (u^2 + 7u + 1)^2$  and  $v = (u^2 - 1) \sqrt{u^2 + 3u + 1} / (u^2 + 7u + 1)^{3/2}$ .

*Proof sketch.* The identity is proved, not fitted, in five steps. [I] By the Schwarzian identity (12),  $w_1 = (dt/d\tau)^{1/2}$  solves  $w'' + Q_V w = 0$ , so  $W$  solves its symmetric square. [II] Elementarily,  $y'' = -Q_V y$  forces  $z = y^2$  to satisfy  $\tilde{N}(z) := z''' + 4Q_V z' + 2Q'_V z = 0$ ; hence  $\tilde{N}(W) = 0$ . [III] Conjugating the hyperkagome operator by the twist unit,  $M_v = v \circ M \circ v^{-1}$  has coefficients in  $\mathbb{Q}(t)$  (the logarithmic derivative  $v'/v$  is rational) and annihilates  $y_0 v$ . [IV] The first-order operator  $T = \rho_0 + \rho_1 d/dt$  satisfies the exact operator identity  $M_v T = Y \tilde{N}$  in the ring  $\mathbb{Q}(t)\langle d/dt \rangle$  of linear differential operators over  $\mathbb{Q}(t)$ , with right remainder identically zero and  $Y$  of order one whose top-order coefficient carries the apparent-singularity polynomial  $p_7$ ; therefore  $T$  maps solutions of  $\tilde{N}$  to solutions of  $M_v$ , and  $T(W)$  solves  $M_v$ . [V]  $T(W)$  is

single-valued near  $t = 0$  ( $W$  and  $W'$  are single-valued functions of  $t$  there, since  $t(q) = q + O(q^2)$  is invertible), and exactly one of the three local solutions of  $M$  at the maximal-unipotent point  $t = 0$  is log-free (Section 5); since  $v(0) = 1 \neq 0$ , conjugation by  $v$  preserves this local structure, so the single-valued solution line of  $M_v$  at  $t = 0$  is one-dimensional and spanned by  $y_0 v$ , and matching a single series coefficient gives  $y_0 v = T(W)$ . As an independent check, (14) is verified as an exact  $q$ -series through order  $q^{120}$ , assembled entirely from the raw  $\eta$  and  $E_2$  definitions above.  $\square$

Three structural remarks follow. (i)  $y_0$  is *quasimodular, not modular*. It is a weight-two *depth-one* quasimodular form on  $\Gamma_0(30)^+$  carrying the orthogonal determinant character of Section 5; equivalently it lives on the genus-one twist double cover  $X(\Gamma_0(30)+2, 3, 6)$  of (8). (ii) *The derivative term is essential*. In the minimal-degree representation (16)—unique, as the fitting nullspace is one-dimensional—the coefficient  $A_1$  is not identically zero: the term  $\rho_1 W'$  is exactly the quasimodular depth introduced by inverting the order-two intertwiner of Section 5. A depth-zero *rational* form  $y_0 v = R(t) W$  is excluded outright by local exponents: at the elliptic point  $t = \frac{1}{9}$  both  $W$  and the twist unit  $v$  vanish to order  $\frac{1}{2}$ , while  $y_0$  carries the exponent  $-\frac{1}{2}$  of  $M$ 's Riemann scheme there (the square-root van Hove singularity,  $[t^m] y_0 \sim 9^m m^{-1/2}$ ), so  $y_0 v$  is regular at  $t = \frac{1}{9}$  whereas  $R(t) W$  has half-integer order for every rational  $R$ . In fact the exclusion is categorical (Theorem 7.3 below): no algebraic factor, at any weight and any level, can absorb the derivative term. This rests on one group-theoretic fact.

**Lemma 7.2.** *The differential Galois group  $\mathcal{G}$  of  $w'' + Q_V w = 0$  over  $\mathbb{C}(t)$  is the full  $\mathrm{SL}(2, \mathbb{C})$ .*

*Proof.* The Wronskian is constant, so  $\mathcal{G} \subseteq \mathrm{SL}(2, \mathbb{C})$ . The solution space of  $\tilde{N}$  is spanned by the products  $w_i w_j$  of solutions of  $w'' + Q_V w = 0$  [32], and by the operator identity of step [IV] the order-one operator  $T = \rho_0 + \rho_1 d/dt$  maps  $\mathrm{Sol}(\tilde{N})$  into  $\mathrm{Sol}(M_v)$ . Any kernel element of  $T$  is a constant multiple of  $z = \exp(-\int \rho_0/\rho_1)$ , and the exact rational computation (with  $r = z'/z = -\rho_0/\rho_1$ )

$$\begin{aligned}
 (18) \quad \frac{\tilde{N}(z)}{z} &= r'' + 3rr' + r^3 + 4Q_V r + 2Q'_V \\
 &= \frac{-30 p_7(t)}{t^2 (t-1)(4t-1)(5t-1)(9t-1)(15t^2 + 17t - 8)^3}
 \end{aligned}$$

—the apparent-singularity polynomial once more, and manifestly nonzero—shows  $z \notin \mathrm{Sol}(\tilde{N})$ . Hence  $T$  is a bijection  $\mathrm{Sol}(\tilde{N}) \rightarrow \mathrm{Sol}(M_v)$ ; being defined

over  $\mathbb{Q}(t)$ , it is equivariant for the differential Galois action, computed in a Picard–Vessiot field containing both solution spaces (whose Galois group restricts *onto* the Galois group of each operator [32]), so it identifies the Galois representation on  $\text{Sol}(M_v)$  with the symmetric square of the standard representation of  $\mathcal{G}$ . Now  $\text{Gal}(M_v)^\circ = \text{Gal}(M)^\circ = \text{SO}(3, \mathbb{C})$  (Section 5): a finite extension of the base field replaces the Galois group by a closed subgroup of finite index, which shares its identity component [32]—applied once to  $M$  and once to  $M_v$  across  $\mathbb{C}(t) \subset \mathbb{C}(t, v)$ —and over  $\mathbb{C}(t, v)$  multiplication by the base-field element  $v$  identifies  $\text{Sol}(M_v) = v \text{Sol}(M)$  equivariantly, with identical matrices. If  $\mathcal{G}$  were a proper closed subgroup of  $\text{SL}(2, \mathbb{C})$ , its identity component would be solvable (it lies in a Borel subgroup or in the normalizer of a torus, or is finite [23, 32]), and then so would the identity component  $\text{Sym}^2(\mathcal{G})^\circ \cong \text{Gal}(M_v)^\circ$  of its symmetric square—contradicting the non-solvability of the simple group  $\text{SO}(3, \mathbb{C})$ . Hence  $\mathcal{G} = \text{SL}(2, \mathbb{C})$ . (Independently, Theorem 6.1 identifies the projective monodromy of  $w'' + Q_V w = 0$  with the lattice  $\Gamma_0(30)^+$ , which is Zariski-dense in  $\text{PSL}(2, \mathbb{C})$ , and the Galois group of a Fuchsian operator is the Zariski closure of its monodromy; either route suffices.)  $\square$

**Theorem 7.3** (No modular-times-algebraic form at any weight). *Let  $F \neq 0$  be a meromorphic modular form of any weight  $k \in \frac{1}{2}\mathbb{Z}$ , with any finite-order multiplier system, on any subgroup  $\Gamma' \subset \text{PSL}(2, \mathbb{R})$  commensurable with  $\Gamma_0(30)^+$ , and let  $a \neq 0$  be any function algebraic over  $\mathbb{C}(t)$ . Then  $y_0 \neq Fa$  (as functions of  $\tau$  under the parametrization (11)). In particular  $y_0$  is not a pure eta quotient, of any level, nor a (weight-two modular form)  $\times$  (algebraic function): every  $\Gamma_0(N)$  and its Atkin–Lehner extensions are commensurable with  $\Gamma_0(30)^+$ , and the eta multiplier has finite order (dividing 24) [12, 26].*

*Proof.* Suppose  $y_0 = Fa$ ; multiplying by the algebraic twist unit  $v$  gives  $y_0 v = F\tilde{a}$  with  $\tilde{a} = av$  still algebraic and nonzero. Work in the Picard–Vessiot field  $K$  of  $w'' + Q_V w = 0$  over  $\mathbb{C}(t)$ : by step [I],  $W$  is a constant multiple of  $w_1^2$ , so  $\mathcal{L} = W'/W = 2w_1'/w_1 \in K$  and the closed form (14) reads  $y_0 v = W(\rho_0 + \rho_1 \mathcal{L})$ . Since  $t$  is  $\Gamma_0(30)^+$ -invariant,  $W \propto dt/d\tau$  is a weight-two meromorphic form with *trivial* multiplier on all of  $\Gamma_0(30)^+$ ; both  $F$  and  $W$  have meromorphic  $q$ -expansions at every cusp of the commensurability class. Pick a positive integer  $n$  such that  $nk \in 2\mathbb{Z}$ —making  $(c\tau + d)^{nk}$  an honest integer-weight automorphy factor, free of the half-integer branch ambiguity, so that the  $n$ -th power of the multiplier system is a genuine finite-order character—and, enlarging  $n$  by the order of that character, such that it is trivial. Then  $\Psi = F^n/W^{nk/2}$  is a weight-zero modular function with trivial multiplier on the finite-index subgroup  $\Gamma' \cap \Gamma_0(30)^+$ , meromorphic at the cusps, hence a rational function



on the compact curve  $X(\Gamma' \cap \Gamma_0(30)^+)$ —a finite cover of  $X(\Gamma_0(30)^+)$ —and therefore algebraic over  $\mathbb{C}(X(\Gamma_0(30)^+)) = \mathbb{C}(t)$ . Raising  $y_0 v = F\tilde{a}$  to the  $n$ -th power, inserting  $F^n = \Psi W^{nk/2}$ , and dividing by  $W^n$  gives

$$(19) \quad (\rho_0 + \rho_1 \mathcal{L})^n = \mathcal{A} W^{n(k-2)/2}, \quad \mathcal{A} = \Psi \tilde{a}^n \neq 0.$$

Every member of (19) lies in  $K$ , so  $\mathcal{A} \in K$  is algebraic over  $\mathbb{C}(t)$ ; because  $\mathcal{G} = \mathrm{SL}(2, \mathbb{C})$  is *connected*, the algebraic closure of  $\mathbb{C}(t)$  in  $K$  is  $\mathbb{C}(t)$  itself [32], so  $\mathcal{A} \in \mathbb{C}(t)$  is fixed by the Galois action. Now act with explicit elements of  $\mathcal{G}$  in the basis  $(w_1, \hat{w})$ , where  $\hat{w}$  is a second, independent solution of  $w'' + Q_V w = 0$ . The torus elements  $\sigma_\lambda : (w_1, \hat{w}) \mapsto (\lambda w_1, \lambda^{-1} \hat{w})$  fix  $\mathcal{L}$  and scale  $W \mapsto \lambda^2 W$ , so applying  $\sigma_\lambda$  to (19) forces  $\lambda^{n(k-2)} = 1$  for every  $\lambda \in \mathbb{C}^\times$ : hence  $k = 2$ . (In particular  $k = 0$  is impossible:  $y_0 v$  is not itself algebraic over  $\mathbb{C}(t)$ .) For  $k = 2$ , (19) reads  $(\rho_0 + \rho_1 \mathcal{L})^n = \mathcal{A} \in \mathbb{C}(t)$ , and the unipotent elements  $\sigma_\mu : (w_1, \hat{w}) \mapsto (w_1 + \mu \hat{w}, \hat{w})$  give  $(\rho_0 + \rho_1 \sigma_\mu(\mathcal{L}))^n = \mathcal{A}$  with  $\sigma_\mu(\mathcal{L}) = 2(w_1 + \mu \hat{w})' / (w_1 + \mu \hat{w})$ . These are pairwise distinct: equal logarithmic derivatives would force the ratio of  $w_1 + \mu \hat{w}$  and  $w_1 + \mu' \hat{w}$  to be a constant of  $K$  (the constants of  $K$  are  $\mathbb{C}$ ), impossible for  $\mu \neq \mu'$  since  $w_1, \hat{w}$  are linearly independent over  $\mathbb{C}$ . Since  $\rho_1 \neq 0$ , the field elements  $\rho_0 + \rho_1 \sigma_\mu(\mathcal{L})$ ,  $\mu \in \mathbb{C}$ , are infinitely many distinct roots in  $K$  of the polynomial  $X^n - \mathcal{A}$ —a contradiction. Hence no such  $F, a$  exist.  $\square$

The quasimodular depth of  $y_0$  is therefore intrinsic—no algebraic factor, at any weight or level, can absorb the  $W'$  term, and  $\rho_1 \neq 0$  is exactly the obstruction. This sharpens consequence (ii) of Section 6 to a categorical exclusion. Both computational inputs (the nonvanishing (18), and, as corroboration, a certified scan excluding any polynomial relation  $P(t, tW'/W) = 0$  through bidegree (28, 12)) are certified in `numerics/certify_y0_lemma.py` (`CERTIFICATE_y0_lemma.txt`). (iii) *Physical and arithmetic fingerprints.* The  $-\frac{1}{2}$  band-edge exponent above is the square-root van Hove singularity of the three-dimensional density of states. The half-integer residues  $\frac{1}{2}$  of  $V_2$ 's first-order coefficient  $a_1$  at  $t = \frac{1}{4}, \frac{1}{5}, \frac{1}{9}$  are the twist multipliers that defeat every trivial-character ansatz, and the quadratic  $15t^2 + 17t - 8$  governing  $\rho_1$  is the one already visible in the rational solution  $R$  of (6). The full symbolic-plus-series certification is provided by the script `numerics/certify_y0.py` (`CERTIFICATE_y0.txt`).

## 8. The generating function itself: an exact obstruction

The closed form (14) concerns the derivative  $y_0 = \Phi'/2$ ; the resolvent sits one integration higher. Since the dispersive part of  $G$  is odd in  $\zeta = z - 1$  with



even central moments  $\nu_m$ , its generating function is exactly

$$(20) \quad \zeta G_{\text{disp}}(z) = \Phi(t), \quad t = \zeta^{-2},$$

term by term at large  $|\zeta|$ . It is natural to ask whether  $\Phi$  itself—hence the Green's function, the flat-band pole being elementary—admits a closed form in the same modular data. It does not, in the following precise sense, and the entire obstruction is one explicit rational function. Integrating (14) once by parts,

$$(21) \quad \Phi(t) = \frac{2\rho_1(t)W}{v} + 2 \int_0^t \Delta(s) \frac{W}{v} ds + \frac{2}{15},$$

$$(22) \quad \Delta := \rho_0 - \left( \rho'_1 - \rho_1 \frac{v'}{v} \right) = \frac{13t-1}{30t(t-1)^2} \neq 0.$$

The identity (21) is proved by differentiation—it reduces to (14) as exact rational-function algebra—with the constant fixed at  $t = 0$ , where  $\Phi(0) = \nu_0 = \frac{2}{3}$  and  $2\rho_1 W/v \rightarrow \frac{8}{15}$ ; the integrand is regular there ( $\Delta W/v \rightarrow -\frac{1}{30}$ ). Note the support of the obstruction:  $\Delta$  has poles only at the cusp image  $t = 0$  and the principal van Hove point  $t = 1$ —the twist locus  $t = \frac{1}{4}, \frac{1}{5}, \frac{1}{9}$  cancels out. If  $\Delta$  vanished, (21) would place  $\Phi$  in the same quasimodular module as  $y_0$ . It does not, and the failure is categorical, not an artifact of this particular rearrangement:

**Theorem 8.1** ( $\Phi$  lies outside the quasimodular module of  $y_0$ ).

$$(23) \quad \Phi \notin \mathbb{Q}(t) + v^{-1}(\mathbb{Q}(t)W + \mathbb{Q}(t)W' + \mathbb{Q}(t)W'').$$

The module on the right is the  $v^{-1}$ -twist of the full  $d/dt$ -closure of  $W$  over  $\mathbb{Q}(t)$  (rank three, by  $\tilde{N}(W) = 0$ ), extended by rational functions; it contains  $y_0$  by (14). The derivative of the generating function is quasimodular; the generating function itself is not.

*Proof.* Three exact steps, certified in `numerics/certify_phi_obstruction.py` (`CERTIFICATE_phi_obstruction.txt`). (a) For any nonzero  $P = S_0 + S_1 d/dt + S_2 d^2/dt^2$  over  $\mathbb{Q}(t)$ , the value  $P(W)$  is neither zero nor algebraic over  $\mathbb{C}(t)$ . Indeed  $P(W) = 0$  would make  $\text{GCRD}(P, \tilde{N})$  a nonconstant right factor of order  $\leq 2$  of  $\tilde{N}$ , which is irreducible: its solution space is the symmetric square of the standard representation of  $\mathcal{G} = \text{SL}(2, \mathbb{C})$  (Lemma 7.2). And an algebraic value is rational, by connectedness of  $\mathcal{G}$  [32];

then  $P$  is constant on the Galois orbit of  $W$ —which spans  $\text{Sol}(\tilde{N})$ , by irreducibility—so  $P$  kills the  $\geq 2$ -dimensional span of orbit differences, again a proper right factor. In particular  $W, W', W''$  are independent over  $\mathbb{Q}(t)$ , and  $v^{-1}\text{span}\{W, W', W''\}$  meets  $\mathbb{C}(t)$  only in 0 (an equality  $\xi/v = r$  would make  $\xi = rv$  algebraic). (b) Suppose  $\Phi = C_0 + (S_0W + S_1W' + S_2W'')/v$  with rational  $C_0, S_i$ . Differentiating, eliminating  $W'''$  by  $\tilde{N}(W) = 0$ , and comparing with  $\Phi' = 2y_0$  coefficient-wise in the independent triple  $W, W', W''$  forces  $C'_0 = 0$  (by the last sentence of (a)),  $S_1 = -d_v S_2$  and  $S_0 = 2\rho_1 + d_v^2 S_2 + 4Q_V S_2$  with  $d_v := d/dt - v'/v$ , and leaves the single condition

$$(24) \quad \tilde{N}_v(S_2) = 2\Delta, \quad \tilde{N}_v := d_v^3 + 4Q_V d_v + 2Q'_V,$$

verified symbolically for generic  $S_2$ . (c) Equation (24) has no rational solution: a rational  $S_2$  could have poles only at  $t = 0, \frac{1}{9}, \frac{1}{5}, \frac{1}{4}, 1$ ; at each of these points the indicial polynomial of  $\tilde{N}_v$  has no negative-integer root and order matching forbids any pole at all, while the analysis at  $t = \infty$  caps the degree at three. The resulting four-dimensional linear system is inconsistent, exactly over  $\mathbb{Q}$ .  $\square$

Physically, by (20),

$$(25) \quad \frac{d}{dz} [\zeta G_{\text{disp}}(z)] = -\frac{4y_0(t)}{\zeta^3}, \quad t = \zeta^{-2} :$$

the object that the closed form (14) evaluates exactly is this derivative combination, while  $\zeta G_{\text{disp}}$  itself lies provably outside the module. The remaining transcendental in (21) is the Eichler-type integral of the weight-four, determinant-character form  $\Delta W^2/v$  (via  $dt = W dq/q$ ); its arithmetic nature—its Eisenstein and cuspidal components on the twist cover of (8)—we leave open.

## 9. The Watson reduction: a generalized body-centered-cubic integral

The orthogonal structure of Section 5 has a concrete origin, which can be exhibited directly. Writing  $\text{CT}_\theta$  for the constant term (torus average) over  $\theta = (\theta_1, \theta_2, \theta_3) \in [0, 2\pi)^3$ , and eliminating the band index of the dispersive bands from the Bloch matrix of the premedial net  $\Gamma$  of Section 2 (the  $K_4$ , or

srs, net) by a resultant, the moment generating function collapses to a single three-dimensional torus integral,

$$(26) \quad \begin{aligned} \Phi(t) &= \frac{2}{3} - \frac{t}{3} \frac{d}{dt} \text{CT}_\theta \log D_+(t, \theta), \\ D_+ &= (1 - 6t + 3t^2) - 2t^2 \sum_i \cos 2\theta_i - 8t^{3/2} \prod_i \cos \theta_i. \end{aligned}$$

The right-hand side of (26) is exactly the structure function of the *body-centered-cubic lattice with first- and second-neighbor hopping*:  $8 \prod_i \cos \theta_i$  is the bcc nearest-neighbor structure function (weight  $t^{3/2}$ ),  $2 \sum_i \cos 2\theta_i$  is the simple-cubic second-neighbor shell (weight  $t^2$ ), and the spectral parameter is  $w(t) = 1 - 6t + 3t^2$ . The hyperkagome LGF is thus expressible through the classical two-parameter “generalized Watson integral”  $I(w; a, b) = \text{CT}_\theta [w - a 8 \prod_i \cos \theta_i - b 2 \sum_i \cos 2\theta_i]^{-1}$  and its parameter derivatives, evaluated along the algebraic curve  $(w, a, b) = (1 - 6t + 3t^2, t^{3/2}, t^2)$ .

Every step is verified in exact arithmetic: the difference-of-squares identity  $\det(I - tS^2) = (1 - 6t + (3 - \Lambda)t^2)^2 - t^3 \Xi^2$  for the Bloch matrix  $S$ ; the 4:1 measure-preserving torus covering  $\kappa = B\theta$  ( $\det B = 4$ ) under which the two invariants become  $\Lambda \mapsto 2 \sum_i \cos 2\theta_i$  and  $\Xi \mapsto 8 \prod_i \cos \theta_i$ ; the cancellation of all half-integer powers of  $t$  in  $\text{CT}_\theta \log D_+$  (so  $\Phi$  is an honest power series in  $t$ ); and the reproduction of the exact moments  $\nu_m$  through  $m = 8$ . Two internal consistencies confirm the picture. The half-integer coupling  $t^{3/2}$  is single-valued only on the  $\sqrt{t}$  double cover—precisely the determinant-character curve  $v^2 = (1 - 4t)(1 - 5t)(1 - 9t)$  of Section 5. And the singular set  $\{\frac{1}{9}, \frac{1}{5}, \frac{1}{4}, 1\}$  is the set of critical values of the bcc(1, 2) band map along the curve, independently reproducing the Riemann scheme (4).

This realization anchors the modular closed form of Section 6 on the lattice side:  $I(w; a, b)$  is the cubic lattice Green's function with second-neighbor interactions, whose natural home is the classical second-neighbor literature (Morita and Horiguchi [28]; the extended Watson integrals of Glasser and Zucker [15]; the cubic modular transformations of Joyce [20])—while the modular structure itself is *proven* in Section 6, independently of any particular literature evaluation. The remaining task is to specialize such an evaluation to the curve above, or to integrate  $D_+$  termwise—it is quadratic in each  $\cos \theta_i$ , so one angular integration yields an inverse square root and the next an elliptic integral, leaving a one-dimensional algebraic period that the orthogonal structure forces into the  $\text{Sym}^2(V_2)$  form. We record the reduction here; matching such a classical evaluation, term by term, against the modular parametrization of Section 6 and the closed form (14) is left to future work.<sup>7</sup>

<sup>7</sup>This reduction is independent of the differential-Galois certificate; it is verified

## 10. Exact results

Independently of the closed-form question, the operator yields exact data. At the symmetric center  $E = 1$  the LGF takes the exact value

$$(27) \quad \operatorname{Re} G(1) = \frac{1}{9},$$

the flat-band contribution  $\frac{1}{3}/(1+2) = \frac{1}{9}$  plus a dispersive principal value that vanishes by the  $E = 1$  symmetry (the value is reproduced numerically by `numerics/verify_specialvalues.py`). (The value at  $E = 2$  is a singular boundary value:  $E = 2$  is the van Hove point  $t = 1$  of (4). The exponents there are  $\{-1, -\frac{1}{2}, \frac{1}{2}\}$ , and the integer-separated pair  $\pm\frac{1}{2}$  permits a logarithm; direct Lorentzian broadening of the Brillouin-zone spectrum (`numerics/verify_vanhove_log.py`) is consistent with a logarithmically divergent  $\operatorname{Im} G$ , and, as the  $E = 1$  symmetry requires, gives the same divergence at  $E = 0$ , which is the same point  $t = 1$ .) From the exponent  $-\frac{1}{2}$  at  $t = \frac{1}{9}$  the band-edge density of states behaves as  $\rho(E) \sim \sqrt{E_{\text{edge}} - E}$ , and the flat band contributes the delta function  $\frac{1}{3}\delta(E+2)$ .

## 11. Conclusion

We have given the exact differential equation that controls the hyperkagome lattice Green's function, determined its differential-Galois structure, proved its modularity, and obtained its weight-two period in explicit closed form. The Green's function is non-Liouvillian—it has no closed form in algebraic or elementary functions—and yet it is modular. The result is also delimited exactly: the derivative  $d[\zeta G_{\text{disp}}]/dz$  is quasimodular through the closed form (14), while the generating function  $\zeta G_{\text{disp}}$  itself provably is not, the entire obstruction being the rational function  $\Delta$  of (22) (Section 8). Constructively,  $M$  is an irreducible third-order Picard–Fuchs operator that is not a *literal* symmetric square, yet whose symmetric square carries a rational solution; its Galois group is therefore the full orthogonal group  $\operatorname{O}(3, \mathbb{C})$ , with identity component  $\operatorname{SO}(3, \mathbb{C})$ , and  $M$  is projectively—through an order-two differential intertwiner—the symmetric square of a second-order operator  $V_2$ . The central result of this paper is that  $V_2$  uniformizes the genus-zero modular curve  $X(\Gamma_0(30)^+)$ , with  $t$  given by the eta-quotient parametrization (11) and the exact Schwarzian identity (12); the projective monodromy of the hyperkagome operator is the arithmetic lattice  $\Gamma_0(30)^+$ . This settles the integral left open by Varma and Monien: the reason a naive elliptic reduction was never in exact arithmetic by `numerics/verify_watson_reduction.py` in the repository.

found is not that the object is genuinely third-order, but that the reduction is hidden behind an operator homomorphism, invisible to the local-exponent tests that detect only function-multiplier and pullback equivalences—and the closed form, once reached, is modular at level 30 rather than elliptic at the classical levels. The Watson reduction of Section 9 anchors the same structure on the lattice side: the LGF is exactly the classical generalized body-centered-cubic integral with first- and second-neighbor hopping, so the classical second-neighbor cubic-lattice literature and the conic/`reduceorder` program of Assis, van Hoeij and Maillard [4] meet the modular parametrization in a single object. The hyperkagome LGF is, to our knowledge, the first three-dimensional lattice Green’s function shown to require such an operator equivalence rather than a literal symmetric-square factorization, and the first realized at a modular level with three distinct prime factors. The weight-two period  $y_0$  itself is exhibited explicitly in Section 7 as a depth-one quasimodular form on  $\Gamma_0(30)^+$  twisted by the determinant character, and is provably not a modular form times an algebraic function at any weight (Theorem 7.3)—in particular, not an eta quotient.

We state the evidential status of these results plainly. Every structural statement above is a rigorous consequence of  $M$  together with Theorems 6.1, 7.1, 7.3 and 8.1; that  $M$  is the *minimal* annihilator of the Green’s function is established to the guess-and-verify standard of the lattice-statistics literature, and an unconditional creative-telescoping proof of minimality remains the one open step.

The result is directly relevant to the modeling of  $\text{Na}_4\text{Ir}_3\text{O}_8$  and related hyperkagome magnets, where the single-particle LGF enters impurity, disorder, and dynamical-mean-field treatments; it provides the exact band-edge and special-point data and a certified equation for high-precision evaluation. More broadly, it adds an explicit three-dimensional frustrated lattice, with a nontrivial orthogonal Galois group, a level-30 modular parametrization, and an explicit depth-one quasimodular closed form for its weight-two period, to the catalog of Picard–Fuchs operators arising in lattice statistics. An unconditional creative-telescoping proof that  $M$  is the minimal annihilator and the analogous Heisenberg/transport Green’s functions are natural next steps.

All lattice constructions, exact moments, the certified operator, and the verification scripts are provided in the public repository accompanying this paper, <https://github.com/bryannasr4-gif/hyperkagome-lgf>.

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